

A ()

A ()

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A ()

i

A

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- Python
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A ():

A

(r, θ)
 $\vec{r}$

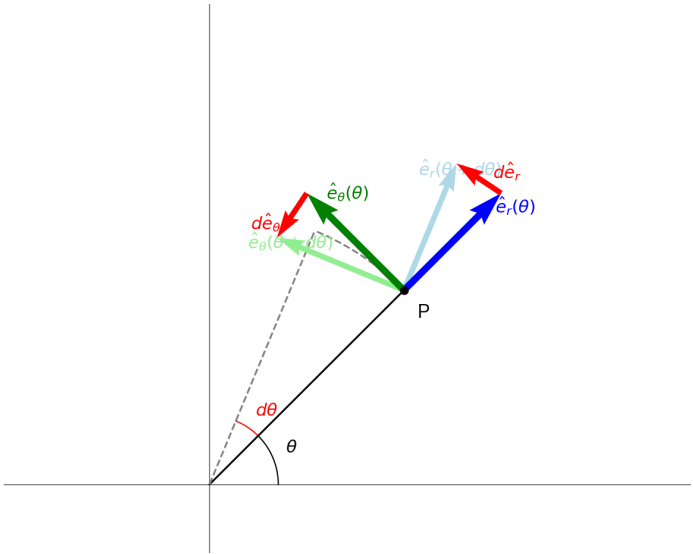
$\hat{e}_r$
 $\hat{e}_\theta$

$(\hat{i}, \hat{j})$
 $\hat{e}_r, \hat{e}_\theta$

θ
 $\theta + d\theta$

$d\hat{e}_r, d\hat{e}_\theta$
 $\vec{r} = r\hat{e}_r$

極座標の単位ベクトルの微分（定性的な図解）



1:

$$\begin{array}{ccccccc}
 & 1. \frac{d\hat{e}_r}{d\hat{e}_\theta} : \hat{e}_r & d\hat{e}_r & \hat{e}_\theta & 1 \cdot d\theta & 2. \frac{d\hat{e}_\theta}{d\hat{e}_r} : \hat{e}_\theta & \\
 d\theta & & & & & & \\
 dt & & & & & & \\
 & & \frac{d\hat{e}_r}{dt} = \dot{\theta}\hat{e}_\theta, & \frac{d\hat{e}_\theta}{dt} = -\dot{\theta}\hat{e}_r & & &
 \end{array}$$


```

//| panel: input
viewof length = Inputs.range([0.1, 2.0], {value: 1.0, label: "    (m)", step: 0.1})
viewof initialAngle = Inputs.range([-Math.PI/2, Math.PI/2], {value: Math.PI/4, label: "    (r"
viewof gravity = Inputs.range([1.0, 20.0], {value: 9.8, label: "    (m/s2)", step: 0.1})

//| echo: false
//| fig-cap: "    "

//    60fps
import { now } from "@observablehq/stdlib"

//
state = {
  const t = now / 1000; //
  //    =  $\sqrt{g/l}$ 
  const omega = Math.sqrt(gravity / length);
  //    (t) = 0 * cos(t)
  const angle = initialAngle * Math.cos(omega * t);

  return {
    time: t,
    angle: angle,
    x: length * Math.sin(angle),
    y: length * Math.cos(angle)
  };
}

```

```
//
width = 400
height = 300
cx = width / 2
cy = 50
scale = 100 // 1m = 100px

// SVG
html`<svg width="${width}" height="${height}" style="background-color: #f8f9fa; border: 1px solid #f8f9fa; border-radius: 10px;">
  <!-- -->
  <circle cx="${cx}" cy="${cy}" r="5" fill="#333" />

  <!-- -->
  <line x1="${cx}" y1="${cy}"
        x2="${cx + state.x * scale}" y2="${cy + state.y * scale}"
        stroke="#666" stroke-width="2" />

  <!-- -->
  <circle cx="${cx + state.x * scale}" cy="${cy + state.y * scale}"
          r="15" fill="#e63946" stroke="#333" stroke-width="2" />
</svg>`
```

l m

$$\frac{d^2\theta}{dt^2} = -\frac{g}{l} \sin \theta$$

$$\theta \ll 1 \quad \sin \theta \approx \theta$$

$$\frac{d^2\theta}{dt^2} = -\frac{g}{l} \theta$$

$$\theta(t) = \theta_0 \cos \left(\sqrt{\frac{g}{l}} t + \phi \right)$$

$$r = ae^{k\theta} \quad (\theta > 0) \quad \dot{\theta} = \omega = \text{const} \quad v_r, v_\theta, a_r, a_\theta$$

$$a > 0 \quad a_r \quad k$$

k

```
///| panel: input
viewof k_param = Inputs.range([-0.5, 0.5], {value: 0.15, label: "    k", step: 0.05})
viewof omega_param = Inputs.range([0.5, 5], {value: 1.5, label: "    ", step: 0.1})
```

```
///| echo: false
import { now } from "@observablehq/stdlib"

//
spiral_state = {
  //
  const raw_t = (now / 1000);
  const t = raw_t % 8; // 8

  const theta = omega_param * t;
  const a_scale = 1.0;
  const r = a_scale * Math.exp(k_param * theta);
  const x = r * Math.cos(theta);
  const y = r * Math.sin(theta);

  return {t, theta, r, x, y, a_scale};
```

```

}

//
canvas_width = 400
canvas_height = 400
center_x = canvas_width / 2
center_y = canvas_height / 2
scale_factor = 20

//
path_points = {
  let pts = [];
  for(let th = 0; th <= spiral_state.theta; th += 0.05) {
    let rad = spiral_state.a_scale * Math.exp(k_param * th);
    // Y
    pts.push(`${center_x + rad * Math.cos(th) * scale_factor},${center_y - rad * Math.sin(th)
  }
  return pts.join(" ");
}

html`<svg width="${canvas_width}" height="${canvas_height}" style="background-color: #f8f9fa
  <!-- X/Y -->
  <line x1="0" y1="${center_y}" x2="${canvas_width}" y2="${center_y}" stroke="#ccc" stroke-d
  <line x1="${center_x}" y1="0" x2="${center_x}" y2="${canvas_height}" stroke="#ccc" stroke-c

  <!-- -->
  <polyline points="${path_points}" fill="none" stroke="#666" stroke-width="2" />

  <!-- -->
  <line x1="${center_x}" y1="${center_y}"
    x2="${center_x + spiral_state.x * scale_factor}"
    y2="${center_y - spiral_state.y * scale_factor}"
    stroke="blue" stroke-width="1.5" opacity="0.5" />

  <!-- -->
  <circle cx="${center_x + spiral_state.x * scale_factor}"
    cy="${center_y - spiral_state.y * scale_factor}"
    r="6" fill="#e63946" />
</svg>`

```

1.

$$\vec{r} = r\hat{e}_r \quad t$$

$$(AB)' = A'B + AB'$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(r\hat{e}_r) = \dot{r}\hat{e}_r + r\dot{\hat{e}}_r$$

$$\dot{\hat{e}}_r = \dot{\theta}\hat{e}_\theta$$

$$\vec{v} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta$$

$$v_r = \dot{r}, v_\theta = r\dot{\theta}$$

$$3 \quad (ABC)' = A'BC + AB'C + ABC'$$

$$\begin{aligned} \vec{a} &= \frac{d\vec{v}}{dt} = \frac{d}{dt}(\dot{r}\hat{e}_r) + \frac{d}{dt}(r\dot{\theta}\hat{e}_\theta) \\ &= (\ddot{r}\hat{e}_r + \dot{r}\dot{\hat{e}}_r) + (\dot{r}\dot{\theta}\hat{e}_\theta + r\ddot{\theta}\hat{e}_\theta + r\dot{\theta}\dot{\hat{e}}_\theta) \end{aligned}$$

$$\dot{\hat{e}}_r = \dot{\theta}\hat{e}_\theta \quad \dot{\hat{e}}_\theta = -\dot{\theta}\hat{e}_r$$

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{e}_\theta$$

$$a_r, a_\theta$$

2.

$$* \quad r = ae^{k\theta} \quad * \quad \dot{\theta} = \omega \quad (\quad) \quad * \quad \ddot{\theta} = \frac{d}{dt}(\omega) = 0$$

$$r \quad \dot{r} \quad \theta \quad t \quad r(t) = ae^{k\theta(t)} \quad t \quad \theta \quad \dot{\theta}$$

$$\dot{r} = \frac{dr}{d\theta} \cdot \frac{d\theta}{dt} = \frac{d}{d\theta}(ae^{k\theta}) \cdot \omega = (ake^{k\theta}) \cdot \omega = ak\omega e^{k\theta}$$

$$ae^{k\theta} = r$$

$$\dot{r} = k\omega r$$

$$\ddot{r} \quad \frac{d}{dt}(C \cdot f(t)) = C \cdot \dot{f}(t)$$

$$\ddot{r} = \frac{d}{dt}(k\omega r) = k\omega \cdot \dot{r}$$

$$\dot{r} = ak\omega e^{k\theta}$$

$$\ddot{r} = k\omega(ak\omega e^{k\theta}) = ak^2\omega^2 e^{k\theta}$$

:

$$v_r = \dot{r} = ak\omega e^{k\theta}$$

$$v_\theta = r\dot{\theta} = (ae^{k\theta})\omega = a\omega e^{k\theta}$$

:

$$a_r = \ddot{r} - r\dot{\theta}^2 = ak^2\omega^2 e^{k\theta} - (ae^{k\theta})\omega^2 = a\omega^2(k^2 - 1)e^{k\theta}$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = (ae^{k\theta})(0) + 2(ak\omega e^{k\theta})\omega = 2ak\omega^2 e^{k\theta}$$

3. $a_r = k$

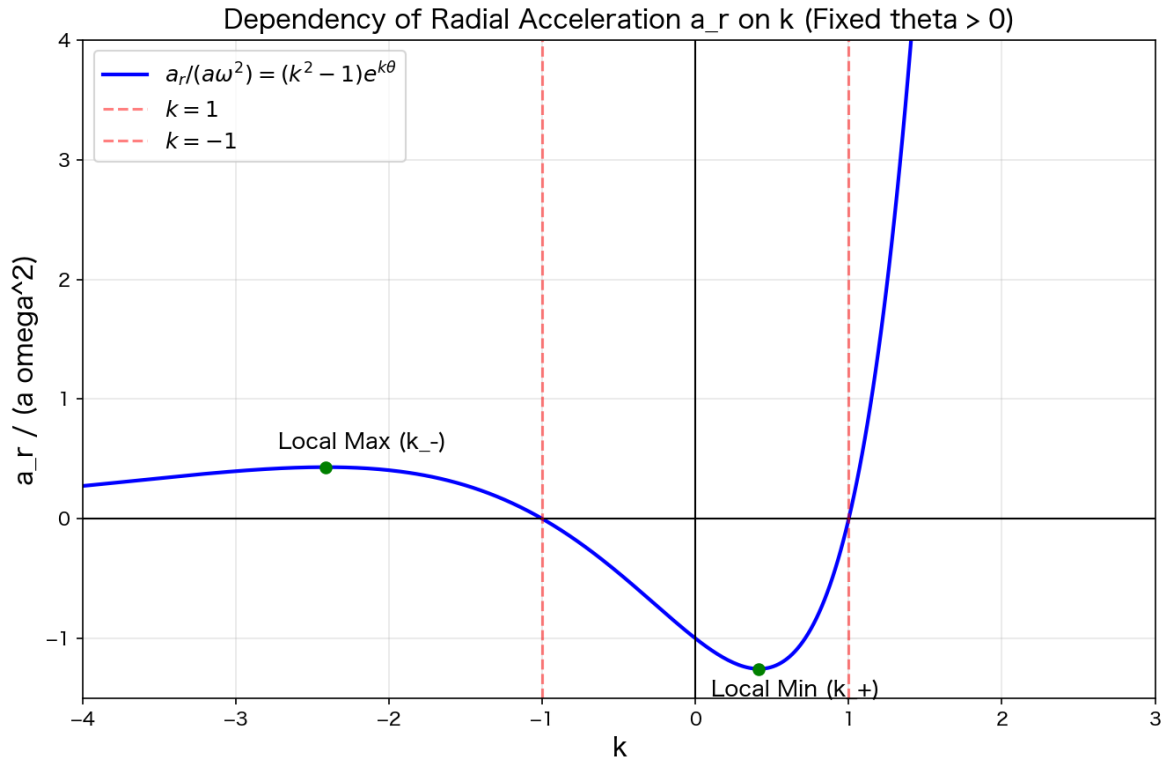
$$\begin{aligned} a_r(k) &= a\omega^2(k^2 - 1)e^{k\theta} & a > 0 & \quad \theta > 0 & \quad k & \quad a\omega^2 > 0 \\ f(k) &= (k^2 - 1)e^{k\theta} \end{aligned}$$

$$f'(k) = 2ke^{k\theta} + (k^2 - 1) \cdot \theta e^{k\theta} = (\theta k^2 + 2k - \theta)e^{k\theta}$$

$$f'(k) = 0 \quad k = 2 \quad \theta k^2 + 2k - \theta = 0$$

$$k_{\pm} = \frac{-1 \pm \sqrt{1 + \theta^2}}{\theta}$$

$$\theta > 0 \quad k_+ > 0 \quad k_- < -1$$



1: a_r k

1. $a_r : e^{k\theta} a_r (k^2 - 1)$

- $|k| < 1 : a_r < 0$
- $|k| > 1 : a_r > 0$
- $k = \pm 1 : a_r = 0$

2. $k \rightarrow \infty : e^{k\theta}$

3. $k \rightarrow -\infty : (k^2 - 1) e^{k\theta} 0 a_r \rightarrow 0 0$

4. $: k < -1 0 < k < 1$